

Computer Vision



Session: 2023 – 2027

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Smart Drinking Waste Detection and Classification

1. Introduction

1.1 Background

The increasing amount of plastic and beverage-related waste has created significant environmental challenges. Proper waste sorting is essential for effective recycling, but manual classification is often time-consuming and prone to errors. Computer vision offers a practical solution by automatically identifying and categorizing different types of waste materials.

1.2 Motivation

The main motivation behind this project is to support smarter and more efficient waste management. By using deep learning techniques for waste classification and detection, the system can assist in reducing manual effort and improve recycling processes.

1.3 Research Questions and Objectives

- Can deep learning accurately classify different types of drinking waste?
- Can object detection identify waste items within an image?
- How can explainable AI help users understand model predictions?

Objectives:

- Develop an automated drinking waste classification system.
- Detect waste objects using YOLOv8.
- Classify detected objects into AluCan, Glass, PET, and HDPEM.
- Explain predictions using Grad-CAM.
- Provide recycling recommendations through an interactive interface.

1.4 Significance of the Study

This study demonstrates how artificial intelligence can be applied to real-world environmental problems. The proposed system supports automated waste sorting, improves recycling efficiency, and highlights the importance of explainable AI in building trust in intelligent computer vision systems.

2. Project Description

This project presents an intelligent computer vision system for automatic drinking waste detection and classification. It combines YOLOv8 for object detection with MobileNetV2 for material classification. The system identifies four drinking waste categories: Aluminum Can (AluCan), Glass, PET, and HDPEM. Grad-CAM is incorporated to explain the model's predictions by highlighting the image regions that contributed most to the decision. A simple rule-based recommendation module provides recycling guidance based on the predicted waste type.

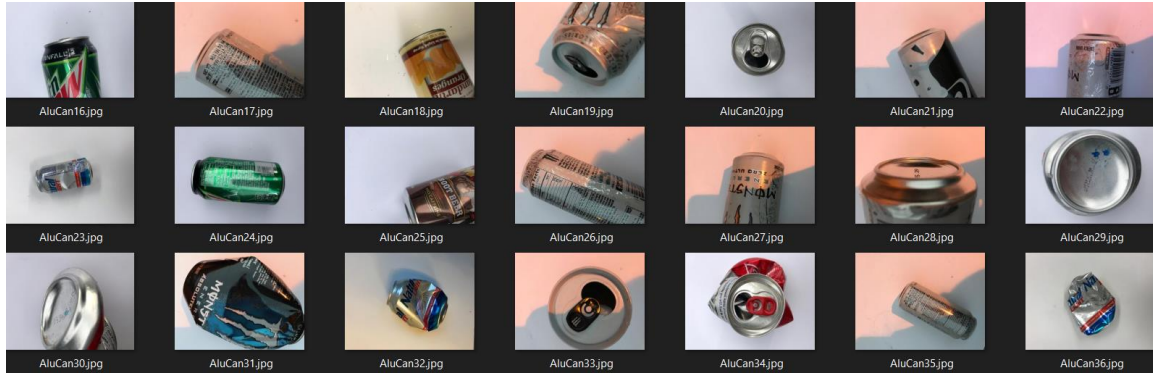
2. Dataset

Dataset: Drinking Waste Classification

Classes:

- AluCan – 1060 images
- Glass – 1232 images
- PET – 1508 images
- HDPEM – 1028 images

Total Images: 4828



A stratified 70%/15%/15% train-validation-test split was used to preserve class distribution. Training images were resized to 224×224 pixels and normalized. Data augmentation included random rotation, translation, zoom, brightness adjustment, and horizontal flipping.

3. Methodology

Step 1: Data preprocessing and augmentation.

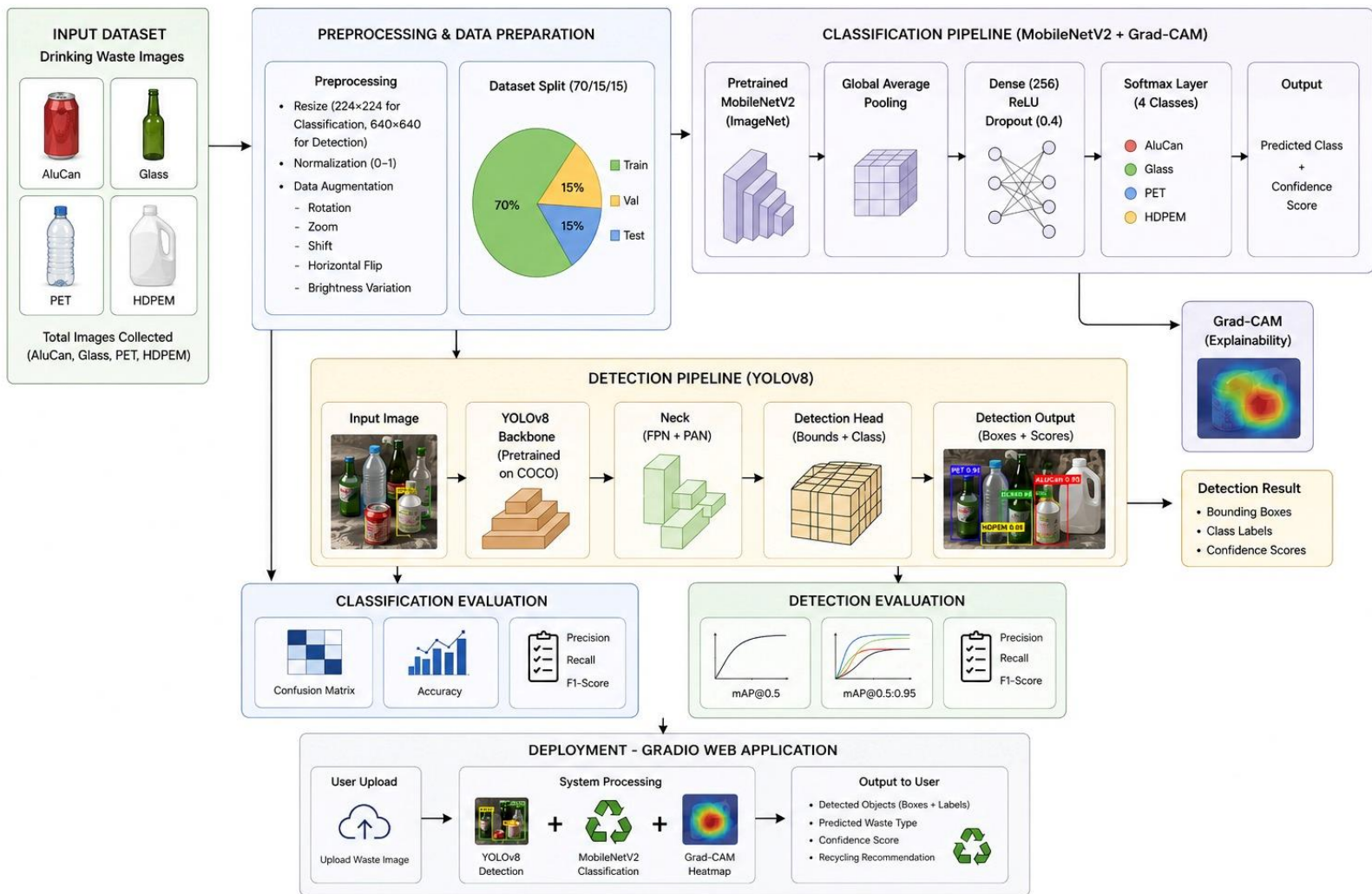
Step 2: YOLOv8 detects waste objects and generates bounding boxes.

Step 3: Each detected object is cropped and passed to a MobileNetV2 classifier.

Step 4: MobileNetV2 predicts one of four material classes.

Step 5: Grad-CAM generates a heatmap explaining the prediction.

Step 6: The user interface displays detection results, classification confidence, Grad-CAM, and recycling recommendations.



4. Model Architecture

4.1 YOLOv8 Detection Model

YOLOv8 was trained to detect drinking waste objects in images. The detector outputs bounding boxes, object confidence scores, and class labels in real time.

4.2 MobileNetV2 Classification Model

MobileNetV2 pretrained on ImageNet was used as the feature extractor. A Global Average Pooling layer, a Dense layer with 256 neurons, Dropout (0.4), and a Softmax output layer were added for four-class classification. Training employed the Adam optimizer, categorical cross-entropy loss, EarlyStopping, ReduceLROnPlateau, ModelCheckpoint, and fine-tuning of the final convolutional layers.

5. Experimental Results and Performance

Metric	Value
Classification Accuracy	96.55%
Precision	96.63%
Recall	96.55%
F1-Score	96.56%
YOLO mAP@0.5	0.9915
YOLO mAP@0.5:0.95	0.8318
YOLO F1-Score	0.9871

Model evaluation included accuracy and loss curves, confusion matrix, precision, recall, F1-score, classification report, and Grad-CAM visualizations. The detector successfully localized waste objects, while the classifier accurately identified material types. Grad-CAM confirmed that the classifier focused primarily on the waste object instead of irrelevant background regions.

6. User Interface

The application integrates YOLOv8, MobileNetV2, and Grad-CAM into a single interface. After uploading an image, the system detects waste objects, classifies each detected object, displays confidence scores, overlays Grad-CAM heatmaps, and provides recycling recommendations based on the predicted material.

7. Conclusion

The proposed framework demonstrates that combining object detection, transfer-learning classification, and explainable AI provides an accurate and practical solution for automated drinking waste sorting. The high classification accuracy and strong detection performance indicate that the system is suitable for smart recycling applications. Future work includes expanding the dataset, supporting additional waste categories, enabling real-time webcam deployment, and optimizing the system for embedded devices.

8. Future Work

- Expand the dataset with additional waste categories.
- Deploy the system on embedded platforms for smart bins.
- Support real-time video processing.
- Investigate more lightweight models for edge deployment.
- Integrate cloud-based monitoring and analytics.